

Coccidioides skin test performance and patterns of prior immunity to coccidioidomycosis among a large population of males entering California state prisons, 2015–2024: a population-based modelling study



Jennifer R. Head,^{a,b,*} Andrew F. Brouwer,^a Arundhati Rajan,^c Marlena Scherer,^d Monica Williams,^e John Balmes,^f George R. Thompson, III^g Neil M. Ampel,^h and Kimberley D. Lucas^d



^aDepartment of Epidemiology, School of Public Health, University of Michigan, Ann Arbor, MI, USA

^bInstitute of Global Change Biology, University of Michigan, Ann Arbor, MI, USA

^cDepartment of Biostatistics, School of Public Health, University of Michigan, Ann Arbor, MI, USA

^dCalifornia Correctional Health Care Services, Public Health Services, Elk Grove, CA, USA

^eCalifornia Correctional Health Care Services, North Kern State Prison, Nursing Services, Delano, CA, USA

^fDivision of Environmental Health Sciences, School of Public Health, University of California, Berkeley, Berkeley, CA, USA

^gDepartment of Internal Medicine, Division of Infectious Diseases and the Department of Medical Microbiology and Immunology, University of California Davis Medical Center, Sacramento, CA, USA

^hCollege of Medicine, University of Arizona, Tucson, AZ, USA

Summary

Background The incidence of coccidioidomycosis—an emerging fungal infection—is high within California state prisons. In 2015, California began offering a *Coccidioides* skin test, Spherusol®, to adult males upon entry into prison, with those susceptible to coccidioidomycosis restricted from placement in two prisons with highest incidence. This study aimed to understand test performance within the incarcerated population and to estimate population-level immunity and force of infection (FOI) by age and pre-incarceration residence.

Methods We fit catalytic models to skin test data from 154,450 males aged 18 to 64 years incarcerated in California state prisons between January 11, 2015, and September 15, 2024.

Findings Estimated test sensitivity was 48.8% (95% confidence interval (CI): 45.1–52.6%). Low sensitivity was attributed to waning reactivity, estimated at 5.3% (95% credible interval (CrI): 4.3, 6.3%) loss in reactivity per year since infection. We estimated a prevalence of prior infection among males from the hyperendemic Kern County of 17.9% (95% CrI: 12.1, 26.0) at 18 years and 94.7% (95% CrI: 89.8, 97.4) by 65 years, and an annual average FOI in Kern County of 54.8 new infections per 1000 susceptible males (95% CrI: 41.6, 70.0). Regions outside the highly endemic area demonstrated lower, but non-negligible, prevalences of prior infection; prior infection prevalence among San Francisco Bay Area male residents reached 16.9% (95% CrI: 13.3, 21.2) by age 65.

Interpretation These findings update our understanding of population immunity to coccidioidomycosis among California males entering prison and inform disease risk in this high-risk population.

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Introduction

Coccidioidomycosis is an emerging infection caused by inhalation of airborne spores from the soil-dwelling fungus, *Coccidioides*. While 60% of infections are

thought to be asymptomatic, symptomatic infection primarily manifests as mild to moderate respiratory disease, with symptoms lasting a median of 120 days,¹ and can become chronic or disseminate to other parts

*Corresponding author. 1415 Washington Heights, Ann Arbor, MI, 48109, USA.

E-mail address: jrhead@umich.edu (J.R. Head).

Research in context

Evidence before this study

Within the southwestern US, incidence of coccidioidomycosis (Valley fever)—an emerging fungal disease—is disproportionately high among residents of California state prisons. In 2015, California began offering a *Coccidioides* skin test upon entry into prison so that non-reactive (susceptible) individuals could be placed outside high-risk facilities. However, little is known about the performance of the skin test nor the patterns of immunity in this population. To assess existing evidence on population immunity to coccidioidomycosis, we searched PubMed from database inception to October 20, 2025, for articles published in English using search terms (“skin test” OR “sensitivity”) AND (“coccidioides” OR “coccidioidomycosis” OR “spherulin” OR “coccidioidin”). We identified 25 studies seeking to establish prevalence of immunity among populations in the Americas. Of these, 19 were conducted prior to 1990 in largely non-representative populations (e.g., school children, outbreak settings), with 12 conducted prior to 1960. All but one used a skin test that has since been discontinued. One large survey conducted in 1949–1951 among students and military recruits established the most recent estimates of population immunity for most California counties. Three studies investigated the sensitivity of the currently available skin test among small ($n = 27$ to 175) populations with confirmed infection within the past 5 years, but none investigated the performance of this test among populations with less recent infection. One study investigated the use of the current skin test among incarcerated residents, but focused on establishing the safety of the test among individuals tested during a single screening event.

Added value of this study

We applied catalytic models to skin test data from 154,450 incarcerated males to estimate population-level immunity and force of infection (FOI) by age and pre-incarceration residence. We documented low test sensitivity among individuals with prior diagnosis of coccidioidomycosis. Our modeling results suggest that low test sensitivity is driven by waning reactivity of the test by time since infection, a phenomenon that is well described for the tuberculin skin test but had yet to be explored for coccidioidomycosis. In all regions, estimated immunity among males entering state prisons exceeded 15% by age 65. Among prior Kern County residents, we estimated 17.9% (95% credible interval (CrI): 12.1, 26.0) were infected by age 18 and 94.7% (95% CrI: 89.8, 97.4) by age 65, with an average annual FOI of 54.8 per 1000 susceptible males (95% CrI: 41.6, 70.0).

Implications of all the available evidence

Skin testing for detecting prior infection with coccidioidomycosis may be unreliable at the individual level due to waning reactivity. This analysis updates our understanding of population immunity to coccidioidomycosis among California males entering prison and yields essential insights needed to evaluate interventions and identify infection risk in high-risk, carceral settings. As a vaccine for coccidioidomycosis is currently under development, understanding natural immunity from infection and how to measure it will be critical for designing coccidioidal vaccine trials.

of the body, causing meningitis and death.² California has seen >8-fold increase in incidence between 2000 and 2018,³ and changing climate conditions threaten to increase the burden of coccidioidomycosis.^{4,5} Incidence rates are disproportionately high among the state's incarcerated populations,⁶ with more than 25% of California state prisons in regions with high incidence of coccidioidomycosis.⁷ Between 1940 and 2015, 11% of reported coccidioidomycosis outbreaks were among incarcerated populations.⁸ In 2017, two prisons, Avenal State Prison and Pleasant Valley State Prison, reported 3959 cases per 100,000 residents, an order of magnitude higher than the highest incidence county in California (Kern County), which reported 311 cases per 100,000 population.⁹

Beginning in 2007, California Correctional Health Care Services (CCHCS) enacted multiple measures to prevent coccidioidomycosis.⁶ One prevention measure, first started in 2015, is a program that screens inmates for infection-derived immunity to *Coccidioides* prior to housing placement.¹⁰ Upon entering reception centers,

individuals are offered (with the option to refuse) a skin test which uses delayed hypersensitivity to a *Coccidioides* antigen, spherulin (Spherusol®; Nielsen BioSciences, Inc.), injected beneath the skin as evidence of prior infection. Only individuals without an overriding medical condition and who either have a history of coccidioidomycosis (according to surveillance data), decline the skin test, or test positive for prior infection by the skin test can be incarcerated at the two prisons with highest incidence: Avenal State Prison (ASP) and Pleasant Valley State Prison (PVSP), both of which are male-only prisons.¹⁰ Skin tests were offered to male residents during a single, mass screening in January 2015 and thereafter were offered to most men aged 18–64 years old upon entry into prison.¹⁰ Skin test information collected by CCHCS represents the largest and most recent source of information on population-level immunity to coccidioidomycosis in California. These data can be used to estimate the probability that an individual entering California state prisons is at risk for the development of coccidioidomycosis, information which is

needed to understand the baseline risk of infection among individuals entering state prisons and to evaluate the effectiveness of the skin test placement program.

Infection with *Coccidioides* results in a cellular immune response which is presumed to result in life-long protection. Detection of delayed dermal hypersensitivity by skin testing is a reflection of the development of coccidioidal cellular immunity.¹¹ However, it has been decades since skin test data were collected at a population level.^{12–18} Data from naval recruits and students in 1949–51 provide the most recent estimates of population immunity for most of California's and Arizona's counties.¹⁴ The skin tests used in these large-scale 1950s studies employed a mycelial-based antigen preparation, coccidioidin. Subsequently, a spherule-based antigen preparation was studied. All coccidioidal skin tests became unavailable in the United States after 1999, and a reformulated test using a spherule-based coccidioidal antigen preparation, spherulin (Spherusol®, Nielsen BioSciences, Inc.), with phenol rather than thimerosal as the preservative, was introduced in 2014.^{19–21} This test has yet to be formally evaluated outside a few, small groups with confirmed history of recent infection.^{20,22,23} These studies found high sensitivity (98%) in a sample of 52 individuals with confirmed, recent history of coccidioidomycosis,²⁰ but lower sensitivities among 27 individuals with active coccidioidomycosis (sensitivity: 55%)²² and 172 individuals with confirmed infection ranging from 0 to 10 years prior (mean: 10 months; sensitivity: 77%).²³ Furthermore, since the 1950s, surveillance records indicate rising incidence and geographic expansion of the disease,³ even as surveillance is believed to vastly undercount infections.²⁴ Measurement of pre-existing anti-coccidioidal immunity is a requirement for the planning and interpretation of vaccine trials,²⁵ identifying and addressing under-diagnoses,²⁴ and characterizing exposures that lead to highest risks of infection.

The objectives of this study are to: 1) evaluate the performance of the Spherusol® skin test among a large population of males incarcerated in California state prisons between 2015 and 2024; 2) estimate age-immunity curves reflecting the probability of prior infection within this population, conditional on age and region of residence; 3) estimate the rate of infection among susceptible individuals within this population—known as the force of infection (FOI)—conditional on age and region of residence.

Methods

Ethics

This study was approved by the Institutional Review Board at the University of Michigan (HUM00241883), with a waiver of informed consent granted. All data were de-identified, and records were linked using a unique ID.

Data sources and description

We obtained information on skin test results, demographic data (age, race, ethnicity), prior history of incarceration, and most recent county of residence for all adult males who were incarcerated in California prisons between January 11, 2015, and September 15, 2024, from the CCHCS electronic health records. Race and ethnicity categories were assigned by CDCR during admission. Skin tests involved injections of 0.1 mL of Spherusol® beneath the skin by a registered nurse.²⁰ Tests were read 48 ± 4 h later and were considered positive for indurations ≥ 5 mm.²⁶

Since 2007, CCHCS has conducted surveillance for coccidioidomycosis within the incarcerated population, with the case definition consistent with that used by the California Department of Public Health (CDPH).²⁷ Prior to January 1, 2019, a confirmed coccidioidomycosis case in California met both laboratory and clinical criteria (illness compatible with coccidioidomycosis) of the Council of State and Territorial Epidemiologists (CSTE) 2011 coccidioidomycosis case definition. Following January 1, 2019, confirmed cases were only required to meet coccidioidomycosis laboratory criteria as defined by CDPH. To identify individuals with known, prior infection for the purposes of test sensitivity calculations, we accessed surveillance records and hospitalization information (recorded from 2012 onward) for all individuals incarcerated any time during the study period. Surveillance records extended before the study period if an individual incarcerated during the study period was diagnosed with coccidioidomycosis during an incarceration period that occurred before January 2015. All individuals incarcerated at their time of diagnosis were recorded in the surveillance system.

Comparison of skin test results by race and ethnicity and pre-incarceration residence

Participants were classified into U.S. census racial and ethnic categories: Hispanic, non-Hispanic (NH) White, NH Black, NH Asian, NH American Indian/Native Hawaiian, and NH Other.²⁸ To examine associations between prior infection and race and ethnicity, we built log-binomial models in which we regressed the binary skin test result against fixed effects for region of most recent residence and race and ethnicity, with a natural spline on age with three roughly equally spaced knots between ages 0 and 65 (ages 16, 32, and 48 years). We computed the prevalence ratio for a given race and ethnicity compared to our reference of Hispanic ethnicity (the most common racial and ethnic group in the study population), controlling for region of pre-incarceration residence and age.

Empirical estimation of test sensitivity

We computed test sensitivity as the probability of a positive skin test among those with a known prior infection, based on record of coccidioidomycosis

diagnosis in the CCHCS surveillance and/or hospitalization database at least 15 days prior to skin test receipt. This cutoff was chosen as a prior study demonstrated lower skin test sensitivity among individuals with active infection, and we examined the robustness of our calculations to this approach in sensitivity analysis (Figure S1).²² To compute 95% confidence intervals on test sensitivity, we regressed a binary indicator for skin test result among those with a prior record of diagnosis in an intercept-only log-binomial model. To understand whether sensitivity varied by age, race, ethnicity, timing of skin test (i.e., during the 2015 mass screening vs. upon intake), immune status, and county of most recent residence, we built log-binomial models that included these variables as fixed effects (Table S1). We examined variation in test sensitivity by time since diagnosis using both quantiles of time since diagnosis (Table S1) and including time since diagnosis in a log-binomial model as a natural spline with four knots, and adjusting for age (Figure S2). Individuals without records of coccidioidomycosis in the CCHCS surveillance database may have had prior asymptomatic or undiagnosed infection, or have been diagnosed with coccidioidomycosis while not incarcerated; therefore, we did not estimate test specificity as we were unable to identify individuals who could be considered, with confidence, to be “true negatives”.

Catalytic estimation of force of infection and age-immunity curves

Catalytic models describe how the proportion of seropositive individuals change with age and are used to estimate the rate at which susceptible individuals acquire an infectious disease.²⁹ At its simplest, for perfectly immunizing diseases, the catalytic model consists of two compartments: an age-dependent susceptible compartment, $S(a)$, and an age-dependent immune compartment, $I(a)$, where individuals remain once infected (Fig. 1A). The relationship between age and prevalence of immunity for an immunizing disease is expected to be monotonically increasing (Fig. 1A). However, when we plotted age vs. the proportion skin testing positive, we observed an inverted U-shaped curve that peaked between ages 50 and 55 years of age and declined thereafter (Fig. 1B). This peak was earlier and more prominent among counties with higher endemicity (e.g., Kern County, Fig. 2). This pattern suggested the possible occurrence of one or two phenomena. First, there could be waning in the ability of the skin test to detect prior infection (i.e., reactivity)—a phenomenon seen for the tuberculin skin test.^{30–33} Second, there could be aging factors that lower test sensitivity among older individuals independent of time since infection—a phenomenon that might occur due to immunosenescence.^{32,33} Accordingly, we adjusted a simple catalytic model by splitting the immune compartment into two compartments

(Fig. 1C): one in which an individual was immune and observable (i.e., true positive test) and one in which an individual was immune and not observable (i.e., false negative test). Susceptible individuals were assumed to move from the susceptible to one of the immune compartments at a rate equal to $\lambda(a) S(a)$. For coccidioidomycosis, the FOI is expected to be higher among working-age adults, who may be engaged in outdoor, higher risk occupations at higher rates.⁸ We thus modelled $\lambda(a)$ using a Weibull distribution as $\lambda(a) = \delta_c \times \text{dweibull}(\alpha; \alpha, \beta)$, where α, β are the shape and scale parameters, respectively, specifying the Weibull distribution and δ_c is a region-specific scaling factor. We classified individuals into a region of most recent residence prior to incarceration. Briefly, all counties outside the highly endemic Southern San Joaquin Valley were categorized into regions based on previous analysis of coccidioidomycosis transmission in California,³ while the five counties within the Southern San Joaquin Valley were defined as their own regions.⁵ There were 13 regions in total (Fig. 2).

Following infection, individuals transitioned to an observable immune compartment (I^o) with a probability equal to test sensitivity among recently infected individuals, denoted $Se_{recent}(a)$, or to an unobservable immune compartment (I^u) with a probability equal to $1 - Se_{recent}(a)$. To account for potential immunosenescence with age, the test sensitivity was allowed to vary by a linear function of age as: $Se_{recent}(a) = Se_{recent,18} - \theta * (a - 18)$, where $Se_{recent,18}$ was the test sensitivity at age 18 immediately following infection, and θ was the loss in test sensitivity among recently infected individuals with each year of age. We then assumed that individuals in the observable immune compartment could move to the unobservable immune compartment at a rate of r , where r represents the exponential rate of waning of the skin test to detect immunity by time since infection. For the sake of simplicity and lack of data on individual exposures, we assumed that there was no immune boosting from re-exposure.

The following system of ordinary differential equations (ODEs) describes movement between the compartments (equation (1)):

$$\frac{dS(a)}{da} = -\lambda(a) \times S(a)$$

$$\frac{dI^o(a)}{da} = \lambda(a) \times S(a) \times Se_{recent}(a) - r \times I^o(a)$$

$$\frac{dI^u(a)}{da} = \lambda(a) \times S(a) \times (1 - Se_{recent}(a)) + r \times I^o(a)$$

We fit the ODEs to data on test positivity to estimate the following 19 parameters: $Se_{recent,18}, \theta, r, \alpha, \beta, \delta_c$ for all 13 regions, and test specificity (Sp , see Supplemental Methods). We defined priors on each parameter

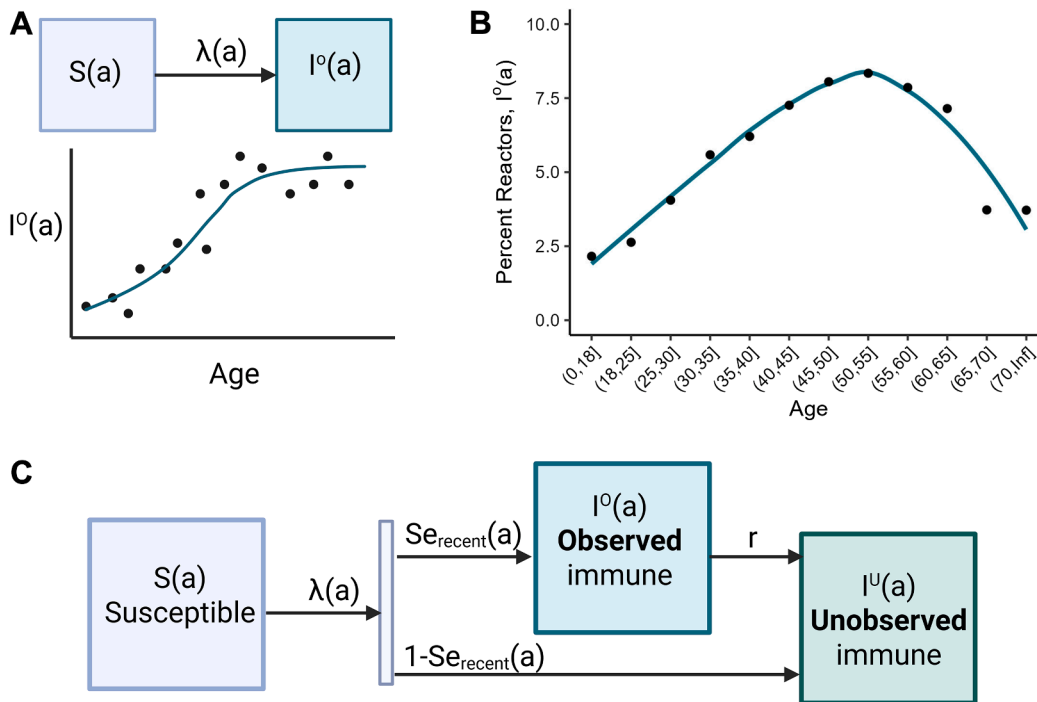


Fig. 1: A) Schematic of a simple catalytic model for an immunizing disease, for which the corresponding age-immunity curve is expected to be monotonically increasing (data in graph are simulated for the sake of demonstration); B) Plot of observed percent of individuals with delayed hypersensitivity ('percent reactors') to the skin test by age bin among the study population. The observed parabolic shape observed suggests potential loss of test sensitivity with time since infection or with age. C) Schematic of the catalytic model used in this study. Individuals are assumed to move from the susceptible, $S(a)$ to the immune class at a rate specified by the force of infection (FOI), $\lambda(a)$, which varies by both age and region of residence. Immediately after infection, the individual is partitioned into either an immune state detectable by the skin test (referred to as "observed immune", $I^0(a)$) or not detectable by the skin test (referred to as "unobserved immune", $I^U(a)$), with partitioning based on age-varying sensitivity ($Se_{recent}(a)$). Individuals in the "observed immune" compartment can enter the "unobserved immune" compartment at a rate of r , which specifies the rate of waning of the skin test to detect prior infection. Diagram created using BioRender.

(Table S2). We used Markov Chain Monte Carlo (MCMC) with Metropolis–Hastings Algorithm to estimate posterior distributions for each of the 19 parameters, accounting for the log likelihood and the log priors. The negative log likelihood (NLL) function was defined as (equation (2)):

$$NLL = \sum_i \left(-k_i^{I^0} \times \log(p_i^{I^0}) - (1 - k_i^{I^0}) \times \log(1 - p_i^{I^0}) \right) + \sum_{j \in \text{"true positives"}} \left(-k_j^{I^0} \times \log\left(\frac{p_j^{I^0}}{p_j^{I^U}}\right) - (1 - k_j^{I^0}) \times \log\left(1 - \frac{p_j^{I^0}}{p_j^{I^U}}\right) \right)$$

This equation is described in greater detail in [Supplemental Methods](#). The first summand reflects information obtained from skin test results of individuals with complete residence data ($n = 102,979$). Briefly, we assumed that the observed skin test positivity for individual i , $k_i^{I^0}$ (1 or 0 based on skin test result), comes from a Bernoulli distribution with

probability of skin test positivity defined by $p_i^{I^0}$, the model-based probability that individual i would test positive by a skin test, given their age and pre-incarceration residence and accounting for both imperfect test sensitivity and imperfect test specificity.

The second summand reflects information obtained on test sensitivity from individuals who had confirmed coccidioidomycosis diagnoses prior to their skin test (e.g., whether the skin test detected those considered to have a true history of infection; $n = 688$) given their age and area of pre-incarceration residence. We assumed that the observed skin test positivity for individual j , $k_j^{I^0}$, comes from a Bernoulli distribution with probability of skin test positivity defined by $\left(\frac{p_j^{I^0}}{p_j^{I^U}}\right)$, the model-

based probability that individual j would test positive by a skin test accounting for test sensitivity and specificity, conditional on the probability that individual j has a true history of infection, both as functions of individual j 's age and area of pre-incarceration residence.

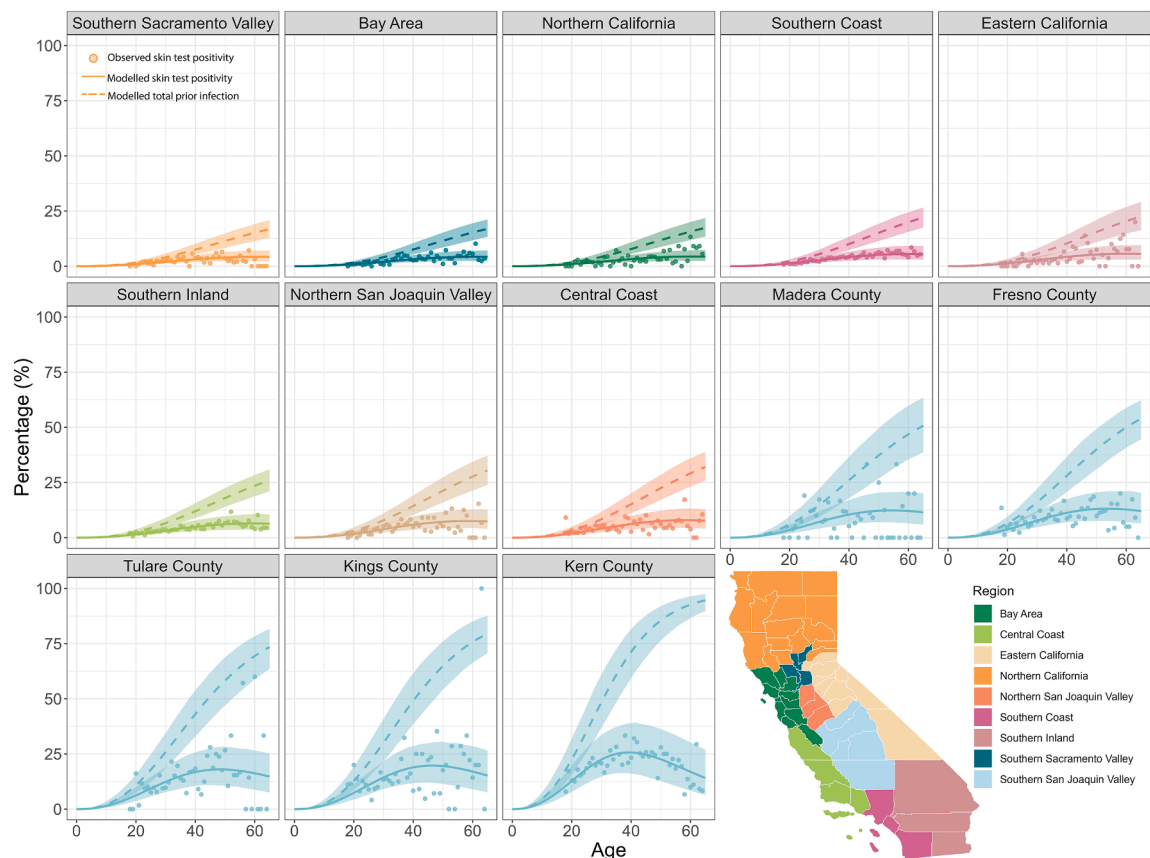


Fig. 2: Estimated percentage of individuals with prior infection with *Coccidioides* among males entering California state prisons by age and region of most recent residence. Dashed colored lines indicate estimated total prevalence of prior infection; solid colored lines indicate estimated prevalence of skin test positivity; shaded colored regions indicate 95% Credible Intervals; colored dots indicate observed (measured) prevalence of skin test positivity. Inset map indicates regions, with all counties in the Southern San Joaquin Valley colored a light blue. [Figure S5](#) presents a zoomed-in visualization of model fit to data by region.

We used the ‘optim’ function in R to identify initial estimates for the parameters. We then ran 1,000,000 MCMC iterations and discarded the first 150,000 as burn-in. We thinned the resulting chain to 0.33% and examined trace and autocorrelation plots for evidence of convergence. To ensure our results were robust to choice of starting conditions, we ran two other chains, adding a 10% jitter in a random direction to the initial set of starting conditions.

To obtain the 95% credible intervals (CrI) on age-FOI and age-immunity curves, we sampled 10,000 parameter sets of the 19 parameters from the joint posterior distribution and calculated the age-FOI and age-immunity curves for each of the 10,000 sets. We took the pointwise 2.5th and 97.5th percentile of those curves as the 95% credible interval.

To arrive at estimates of population-weighted mean FOI and mean immunity within regions, we obtained information on the age distribution of males aged 18–64 in California counties using the 2020 decennial

census and accessed using the tidycensus package in R.³⁴ We then took a weighted average of the age-specific estimates for FOI and immunity across the age range, weighting by region population per age. Accordingly, all mean estimates of immunity and FOI can be interpreted as the mean prevalence of immunity or mean FOI among males entering prison if they had the same age distribution as the general population within the region of residence. All analyses were conducted in R version 4.3.0.³⁵

Role of the funding source

Funders had no role in study design, data collection, data analysis, interpretation, writing of the report or decision to submit.

Results

Descriptive statistics

There were 1,264,325 individuals incarcerated in California male state prisons for at least one day between

January 11, 2015, and September 15, 2024. During this period, the skin test was offered to 309,980 distinct individuals. Of these individuals, 154,450 (49.8%) accepted the skin test, including 37,192 individuals who received the test during the one-time mass screening (Jan 11, 2015) and 117,258 individuals who received the test after the mass screening, during intake. After adjusting for age, race/ethnicity, region of prior residence, history of incarceration, and timing of skin test, we estimated that older individuals were more likely to accept the test as compared to younger individuals and individuals were more likely to accept the test upon admittance than during the mass screening (Table S3). NH Black individuals were least likely to accept the test compared to other racial/ethnic groups. Compared to individuals who were screened upon intake, individuals screened during the one-time screening had longer histories of prior incarceration (median of 1631 days vs. 168 days) but were demographically similar (Figure S3). Information on the region of most recent pre-incarceration residence was available for 102,979 tested individuals, of whom 8401 (5.4%) had a positive skin test. Test positivity varied by region of most recent pre-incarceration residence and was highest among counties in the Southern San Joaquin Valley (Kern: 1059/5079, 20.9%; Kings: 196/1393, 14.1%) and lowest in Northern regions (San Francisco Bay Area Region: 213/8534, 2.5%; Southern Sacramento Valley: 159/6457, 2.5%; Northern California: 137/4882, 2.8%; Table S4).

Differences in test positivity by race and ethnicity

Information on race and ethnicity was available for all tested individuals. Test positivity varied by race and ethnicity: NH American Indian/Native Hawaiian (111/1875; 5.9%), NH White (2186/38,004; 5.8%), Hispanic (4337/76,249; 5.7%), NH Black (1423/30,225; 4.7%) and Asian (74/2506; 3.0%; Table S4). Compared to individuals with Hispanic ethnicity, prevalence of skin test positivity was significantly lower among individuals who were NH Asian (prevalence ratio (PR): 0.55, 95% CI: 0.40, 0.74), NH Black (PR: 0.86, 95% CI: 0.80, 0.93), and NH Other (PR: 0.66, 95% CI: 0.53, 0.81). Compared to individuals with Hispanic ethnicity, prevalence of skin test positivity was not significantly different among individuals who were NH White (PR: 0.96, 95% CI: 0.90, 1.02) or NH American Indian/Alaska Native (PR: 1.16, 95% CI: 0.91, 1.44).

Empirical test sensitivity

Among the 154,450 individuals tested, 688 had a record of coccidioidomycosis diagnosis in the CCHCS database that occurred earlier than 15 days prior to their test. Of these, 336 had a positive skin test result, corresponding to an overall test sensitivity of 48.8% (95% confidence interval (CI): 45.1–52.6%). Sensitivity was higher during the initial screening in January 2015

(168/282; 60%, 95% CI: 54–66%) than upon intake (168/406; 41%, 95% CI: 37–46%) and lower among individuals considered high medical risk, either due to an immunocompromising or other condition (e.g., diabetes, older age; Table S1). Test sensitivity was highest approximately 2–10 years following diagnosis, and dropped thereafter (Figure S2, green line). Results were robust to the choice of coccidioidomycosis diagnosis 15 days prior to the test as the cutoff for considering individuals “true positives” (Figure S1).

Modelled estimates of test performance, force of infection and age-immunity curves

Model assessment indicated convergence of the MCMC chains (Figure S4a–e) and model output indicated overall good fit of the observed immunity curve to the observed data from the study cohort (Figure S5; Fig. 2). In our MCMC models, we estimated that the sensitivity of the test in a recently infected individual at age 18 ($Se_{recent,18}$) was 75.7% (95% credible interval (CrI): 66.2, 86.2; Table S2). We estimated a decline in test sensitivity immediately following infection (θ) as 0.2 percentage points per year of age (95% CrI: 0.1, 0.6), suggesting little loss of test sensitivity due to immunosenescence. We estimated an annual rate of waning of test sensitivity over time since infection to be $1 - \exp(-r) = 5.3\%$ per year (95% CrI: 4.3, 6.3%), meaning that each year, 53 of every 1000 individuals with prior infection would become skin test negative (Table S2). After computing the probability of true infection and observed skin test positivity for each individual in the study population, conditional on age and pre-incarceration residence, we compared the modelled test sensitivity by time since infection to the empirically estimated test sensitivity (Figure S2). Using estimated model parameters, we calculated that we would expect to observe a sensitivity of 40.1% (95% CrI: 39.2, 41.0) among the study population. This model-estimated population test sensitivity slightly underestimates the empirically estimated test sensitivity (48.8%).

The population-level mean FOI per region ranged from 3.5 (95% CrI: 2.5, 4.5) annual infections per 1000 susceptible males with recent residence in the Southern Sacramento Valley region to 54.8 (95% CrI: 41.6, 70.0) in Kern County (Table 1; Fig. 3A). We estimated an FOI curve that rose from birth until a maximum of 49 years, after which it declined, with the FOI at age 65 remaining higher than the FOI at age 20 (Figure S6). At its peak, the age-varying FOI in Kern County reached 70.3 infections per 1000 susceptible individuals aged 49 years old (95% CrI: 55.6, 85.4; Fig. 3B).

The highest immunity was estimated from Kern County, where a mean of 60.5% (95% CrI: 51.1, 69.6) of the adult male population was estimated to have prior infection with *Coccidioides* (Table 1; Fig. 4A). Among males with recent residence in Kern County, we estimated that 17.9% (95% CrI: 12.1, 26.0) of 18-year-olds,

Region/County	Population immunity (per 100 population)				Force of infection (per 1000 susceptible individuals)			
	Adult males, 18–65 years	Age-stratified			Adult males, 18–65 years	Age-stratified		
		18 years	49 years ^a	65 years		18 years	49 years ^a	65 years
Southern Sacramento Valley	7.5 (5.4, 9.9)	1.2 (0.8, 1.9)	11.0 (8.3, 14.5)	16.8 (12.8, 20.8)	3.5 (2.5, 4.5)	1.7 (1.1, 2.4)	4.4 (3.4, 5.5)	3.8 (2.9, 4.5)
Bay area	7.9 (5.8, 10.5)	1.2 (0.8, 1.9)	11.1 (8.3, 14.8)	16.9 (13.3, 21.2)	3.6 (2.7, 4.7)	1.7 (1.1, 2.5)	4.4 (3.4, 5.7)	3.8 (2.9, 4.6)
Northern California	8.3 (6.1, 11.1)	1.3 (0.8, 2.0)	11.4 (8.2, 15.0)	17.4 (13.4, 21.8)	3.7 (2.7, 4.8)	1.7 (1.2, 2.6)	4.6 (3.4, 5.7)	3.9 (3.0, 4.7)
Southern Coast	10.0 (7.5, 12.9)	1.7 (1.1, 2.5)	14.6 (11.3, 18.8)	22.0 (17.5, 26.5)	4.7 (3.6, 6.0)	2.3 (1.5, 3.2)	6.0 (4.7, 7.3)	5.1 (4.0, 6.0)
Eastern California	11.6 (8.3, 15.8)	1.7 (1, 2.7)	15.0 (10.5, 20.4)	22.6 (16.4, 29.1)	5.1 (3.5, 6.9)	2.3 (1.5, 3.4)	6.1 (4.3, 8.2)	5.2 (3.6, 6.8)
Southern Inland	11.7 (8.2, 15.8)	2 (1.3, 3.1)	17.4 (13.7, 22.5)	25.9 (21.1, 30.9)	5.7 (4.4, 7.2)	2.7 (1.9, 3.9)	7.2 (5.7, 9.0)	6.1 (4.9, 7.3)
Northern San Joaquin Valley	13.9 (10.3, 18.1)	2.4 (1.5, 3.8)	20.6 (15.4, 26.7)	30.5 (23.9, 37.2)	6.9 (5.0, 9.0)	3.3 (2.2, 4.8)	8.7 (6.6, 11.0)	7.4 (5.7, 9.1)
Central Coast	14.9 (11.2, 19.2)	2.6 (1.7, 3.9)	21.7 (16.6, 28.2)	32.0 (25.9, 38.8)	7.3 (5.5, 9.5)	3.5 (2.3, 5.0)	9.2 (7.2, 11.5)	7.9 (6.2, 9.6)
Madera County	25.2 (17.9, 34.3)	4.6 (2.7, 7.5)	36.2 (25.9, 48.3)	50.7 (38.7, 63.5)	13.4 (9.0, 19.4)	6.4 (4.0, 10.1)	17.0 (11.6, 24)	14.5 (9.9, 20.1)
Fresno County	26.0 (19.9, 32.8)	5.1 (3.2, 7.6)	38.9 (30.5, 47.4)	53.9 (44.7, 62.2)	14.6 (10.8, 18.9)	7.0 (4.8, 10.2)	18.6 (14.3, 22.9)	15.9 (12.4, 19.1)
Tulare County	38.6 (30.3, 47.2)	8.6 (5.6, 12.9)	56.9 (46.3, 68.9)	73.5 (63.6, 81.7)	24.9 (18.4, 32.6)	12.0 (8.1, 17.3)	31.8 (24.3, 40.7)	27.1 (20.9, 33.4)
Kings County	42.0 (33.5, 52.1)	10.1 (6.7, 15.6)	63.5 (52.6, 74.5)	79.6 (70.8, 87.7)	29.5 (22.0, 40.0)	14.4 (9.7, 21.5)	38.0 (29.4, 48.6)	32.5 (25.2, 41.1)
Kern County	60.5 (51.1, 69.6)	17.9 (12.1, 26.0)	84.4 (76.3, 91.4)	94.7 (89.8, 97.4)	54.8 (41.6, 70.0)	26.6 (18.2, 38.6)	70.2 (55.8, 85.4)	59.9 (47.5, 70.7)

^aIndicates the age at which the force of infection peaks.

Table 1: Estimated values of population immunity (per 100 population) and force of infection (per 1000 susceptible individuals) among males entering California state prisons between 2015 and 2025, by age and most recent region of residence.

84.3% (95% CrI: 76.3, 91.4) of 49-year-olds, and 94.7% (95% CrI: 89.8, 97.4) of 65-year-olds had a prior infection. The second highest immunity was estimated among those with prior Kings County residence (Table 1; Fig. 4A), where the prevalence of prior infection was estimated at a mean of 42.0% (95% CrI: 33.5, 52.1) and at 10.1% (95% CrI: 6.7, 15.6), 63.2% (95% CrI: 52.6, 74.5), and 79.6% (70.8, 87.7) among individuals aged 18, 49, and 65 years of age, respectively. Less endemic counties still had non-negligible prevalence of prior infection. For instance, the prevalence of prior infection among prior San Francisco Bay Area region residents was estimated at a mean of 7.9% (95% CrI:

5.8, 10.5) and reaching a prevalence of 16.9% (95% CrI: 13.3, 21.2) by age 65. Parameter estimates were consistent across different starting conditions (Figures S7 and S8).

Discussion

In a large population of adult males incarcerated in California state prisons, we documented low test sensitivity of the Spherosol® skin test for coccidioidomycosis and estimated age- and region-stratified estimates of the prevalence of infection-derived immunity to *Coccidioides*. This study reveals important

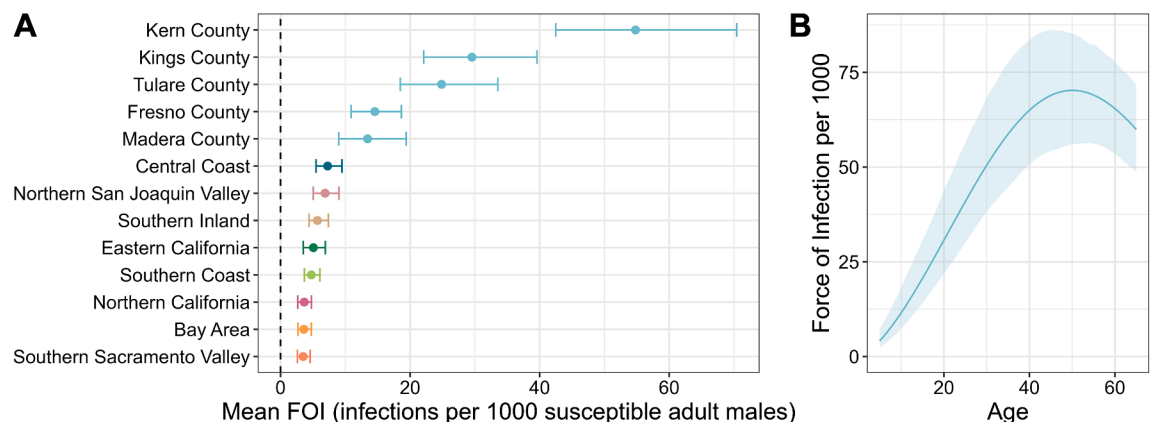


Fig. 3: Estimates of annual force of infection (FOI), defined as the rate at which susceptible individuals become infected. A) Mean FOI among susceptible adult males aged 18–65 years by region of residence. Light blue colors all indicate counties within the Southern San Joaquin Valley; B) Estimated age-FOI curve for Kern County, demonstrating variation in FOI over the life-course.

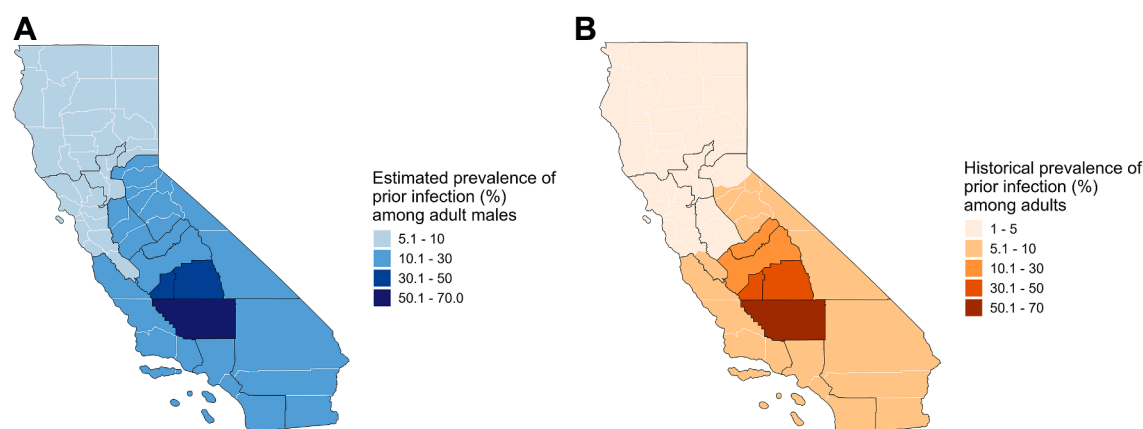


Fig. 4: Map of estimated prevalence of prior infection with *Coccidioides* among A) adult males entering California state prisons between 2015 and 2024 and B) adult naval recruits and students in 1949–1951 who were lifelong residents of their counties (adapted from Edwards and Palmer, 1957¹⁴). The ability to directly compare prevalence over time is limited by differences in diagnostic antigens used and the populations surveyed.

information about the performance of the Spherusol® skin test (reformulated in 2014 by Nielsen Biosciences, Inc.), which to date, had not been formally evaluated outside small, controlled settings.^{20–22} Prior work evaluating the Spherusol® skin test estimated a test sensitivity of 98.3% among 52 individuals with a recent history of pulmonary coccidioidomycosis.²⁰ Using skin test data from prior coccidioidomycosis cases, we estimated a lower test sensitivity of 48% in our study population. Our model results suggested a test sensitivity of approximately 76% among 18-year-old individuals with recent infection, with lower sensitivity among individuals whose prior infection may not have been recent. While waning of skin test reactivity has not been formally documented for the *Coccidioides* skin test, historical tests using an earlier formulation documented a flattening or decline in the prevalence of reactive skin tests with age after 12–15 years of residence in an endemic location.^{12,36,37} Lack of experimental data on mechanisms underpinning the loss of reactivity to the Spherusol® skin test motivates future study. However, progressive loss of reactivity with age to the tuberculin skin test is well documented and reflects a combination of general changes in the cellular immunologic mechanism, local factors in the aging skin, and a gradual loss of reactivity with time since infection.^{32,33} We estimated a waning of the Spherusol® skin test to detect prior infection at a rate of approximately 5.3% per year since infection. At the same time, our models estimated low loss in test sensitivity per year of age, holding time since infection constant; this phenomenon would have been consistent with patterns of immunosenescence that occur among older adults, resulting in less robust recruitment of T cells to the site of injection.^{32,33} Due to waning reactivity to tuberculin by time since infection, tuberculin skin test guidelines

recommend that an individual who tests negative by the tuberculin skin test should receive a second test 1–3 weeks later to confirm.³⁸ Among individuals with a prior infection to tuberculosis, the initial injection can help “boost” specifically sensitized T lymphocytes, which are then recruited during the second test.³¹ Based on our model-derived estimates for initial sensitivity and waning of sensitivity by age and time, we would expect a 40% test sensitivity, which is slightly lower than the empirically estimated 48%. Therefore, it is possible that our estimate for test sensitivity following infection is a slight underestimate and/or our estimate for loss of test sensitivity by time since infection is a slight overestimate.

Immunity to *Coccidioides* is believed to be lifelong even as reactivity to the Spherusol® skin test may wane; thus, the skin test may not be a reliable indicator of immunity at the individual level. Our findings therefore motivate follow-on analysis of the effectiveness of the skin test placement program in averting cases of coccidioidomycosis in prisons. Nevertheless, our finding of high probability of a false negative test suggests that the current risk-stratified placement strategy at least errs on the side of caution (i.e., in that it favors restriction of individuals from prisons with the highest coccidioidomycosis incidence). Estimating the probability of a false positive test is needed to confirm whether the placement strategy is truly conservative. Beyond estimation of test specificity, a better understanding of whether offering a second test 1–3 weeks following a negative test, as recommended for tuberculosis, would improve accuracy is needed, and if so, whether a more accurate risk stratification approach would justify the additional cost and workload associated with such a strategy. For tuberculosis, an interferon gamma release assay (IGRA) offers an alternative

means of assessing cell-mediated immunity, using a blood sample; alternative approaches to assessing cell-mediated immunity to *Coccidioides*, including using an ex vivo cytokine release assay, could similarly be considered.³⁹

This study provides insights into the true burden of coccidioidomycosis in California as well as shifts in burden over time, which is important as surveillance systems undercount cases.²⁴ Our estimated population-level mean FOI suggests that there would be nearly 5480 new infections (including asymptomatic infections) per 100,000 susceptible adult males entering prison in Kern County on average, per year. Surveillance data from 2021 indicate a reported incidence rate of 306.2 per 100,000 individuals (regardless of age and sex) in Kern County.²⁴ Assuming that 40% of infections are symptomatic,² a rough “back-of-the-envelope” calculation suggests that, if the estimates for the prison population were to be generalized to the entire county (albeit with caution, see limitations), about 1 in 7.5 symptomatic infections would be captured by surveillance and 1 in 18 overall infections would be captured. This aligns with prior literature estimating that 1 in 5 to 1 in 8 symptomatic infections in California and Arizona are captured by surveillance.⁴⁰ Furthermore, we estimated a mean immunity of 14% among individuals with prior residence in the Northern San Joaquin Valley region (Table 1, Fig. 4A), where incidence has been increasing since 2000,³ higher than the estimate of 1–5% during the 1949–1951 surveys (Fig. 4B). We note that direct comparisons between 1949 and 1951 estimates and our estimates are limited due to differences in diagnostic antigens used (coccidioidin vs. spherulin); differences in the residence history, demographics, and occupations of study participants; and potential changes in climate suitability for *Coccidioides* over time.

This study has limitations. First, we were unable to empirically estimate test specificity in this cohort as we could not reliably categorize individuals as “true negatives” based on the data provided. However, test specificity as reported by the manufacturer exceeds 98%,²⁰ and internal estimates based on the initial mass screening skin test data from this population suggest that the specificity is indeed around 97%.⁴¹ A small number of patients (2%) with histoplasmosis had cross-reactions when spherulin was used in the skin test.⁴² However, this was not seen with Spherusol®, where twelve subjects with a recent history of pulmonary histoplasmosis were skin test negative.²⁰

Second, the empirically estimated test sensitivity for our population may not be generalizable to other populations. The higher test sensitivity during the initial mass screening may have been due, in part, to longer histories of incarceration among that subgroup (Figure S3). As incarceration appears to be a risk factor for infection, prior infections among the already incarcerated subpopulation may have been more recent

than among the subpopulation tested upon entry. Other unmeasured differences or cohort effects between the population tested in the 2015 mass screening and the population tested upon admission may further explain differences in test sensitivity, even as the two populations were similar based on measured demographics, and incidence in 2012–2015 was relatively low, concurrent with drought conditions in the state.⁴³ Individuals incarcerated in prisons tend to be older than individuals incarcerated in jails, so a similar program implemented for jails may yield potentially higher test sensitivity. Finally, we used the date of diagnosis to estimate the relationship between empirical test sensitivity and time since infection. Although we anticipate faster diagnosis of coccidioidomycosis within the carceral setting, delays in diagnosis may be highly variable and lengthy, with enhanced surveillance from Arizona estimating a median of 55 days between disease onset and diagnosis.¹

Third, our estimates for population immunity and FOI may not be generalizable to other adult populations in California. Although we adjusted for differential age distributions between the general California male population and the population entering state prisons when computing the population-level mean FOI and mean immunity among adult males by most recent residence before incarceration, individuals in our study population may differ from the general California population in other ways. First, prior incarceration may have contributed to higher rates of prior infection than would be otherwise be observed among the general population. Although the majority of individuals in our study population were tested upon entry into state prisons, 45% had a previous history of incarceration. Additionally, we simplified our model by assuming that there is no immune-boosting effect of repeated exposures over time, but there has been limited research on this phenomenon. Because incarceration is a risk factor for exposure to *Coccidioides*, individuals with a history of incarceration may have had boosted immune profiles. Future work may aim to understand differences in prior infection based on duration and location of prior incarceration. Second, individuals who become incarcerated may differ from individuals who never become incarcerated on a variety of factors potentially related to infection status. For instance, they may have higher rates of immunocompromising conditions, which could contribute to higher infection rates and lower test sensitivity among this population. Differential test acceptance rates by race/ethnicity and during the mass screening vs. upon intake could lead to slight biases in estimated immunity even as stratification of findings by age and region of prior residence guards against bias due to differential test acceptance across age and residence. Finally, our findings are likely not generalizable to females, who have lower incidence rates for coccidioidomycosis in California.

Fourth, we captured information on region of most recent residence prior to incarceration but lacked information on other relevant pre-incarceration exposures, including occupation, duration of time in most recent residence, lifetime residential history, or travel history. We are therefore unable to confirm whether individuals with positive skin tests acquired infection at their prior residence, while traveling to an endemic location, or during a prior incarceration period. Prevalence of immunity among lifetime residents of highly endemic counties may therefore be higher than our estimates. However, our estimates for the mean prevalence of immunity among adult males in Kern County was 61%, which falls within the range of estimated immunity of 50–70% measured during the 1949–1951 surveys of individuals who had lived in Kern County their whole life (Fig. 4), even as other discussed reasons for systematic differences between the two prevalence estimates exist.

Conclusions

The Spherusol® skin test for coccidioidomycosis may not be a reliable indicator of immunity at the individual level due to potential waning of reactivity. This study is important for the health of incarcerated individuals as it can be used to characterize the baseline risk of coccidioidomycosis among individuals entering state prisons based on their age and prior residence. Further, this work lays the foundation to enable evaluation of the effectiveness of the skin test program at averting infections by informing placement decisions. Finally, this analysis represents the largest study on population immunity to coccidioidomycosis since the 1950s, laying the groundwork to enable estimation of the true burden of coccidioidomycosis in California amidst high case under-ascertainment.

Contributors

JRH and KDL conceptualized the study. MS, MW and KDL curated the data. JRH and AR performed data analysis and produced visualizations. JRH, AFB, JB, GRT, and NA assisted with methodology. JRH procured funding and wrote the original draft. All authors reviewed and edited the manuscript and accept responsibility for submission. JRH and KDL have directly accessed and verified the underlying data reported in the manuscript and made the decision to submit the manuscript.

Data sharing statement

Data were made available via a data use agreement with the California Correctional Health Care Services. Code used in the analysis is available at <https://github.com/jrhead/CocciSIModel>.

Editor disclaimer

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Declaration of interests

In the 36 months prior to publication, GRT received consulting fees from Astellas, Basilea, Cidara, Elion, F2G, and Scynexis and payment for participation in an advisory board to Pfizer. JB received grants from the NIH, US EPA, and California Energy Commission, payment for being a Physician Member of the California Air Resources Board, and

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.lana.2026.101461>.

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